

SAM SEIFI

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Education

- BSc - Physics with Astrophysics and Cosmology**, King's College London Sept 2025 - exp. 2028
- **Year 1 average: 64%** — **Computing 91%**, Laboratory 72%, Astrophysics 67%, Mathematics 65%
- General Certificate of Education (GCE A-level)**, The Compton Sixth Form Sept 2023 - Aug 2025
- **Achived Grades: ABB** | Mathematics, Physics and Computer Science

Personal Projects — Additional Works: svsam.com

- EvoXRB: Genetic Algorithms, and a Very Real Black Hole** | [↗](#)
- Building a Python, NICER-inspired X-ray experiment around MAXI J1820+070, then finding out whether natural selection is actually any good at fitting spectra. The project ended up forcing me to learn far more than the algorithm. [Read more at the blog post here](#) | [↗](#).
- Exoplanet orbital temperatures and modelling eccentricity using training data** | [↗](#)
- Python simulation and analysis pipeline using NASA Exoplanet Archive data to replicate exoplanet orbital dynamics and estimate orbital thermal variation.
 - Generated orbital-temperature profiles across a full orbital period, with each exoplanet selectable for simulation and comparison.
- Desktop Macro** | [↗](#)
- Local desktop macro recorder inspired by TinyTask. It records mouse movement, clicks, scrolling, and keyboard press-and-release timing only while recording is visibly enabled, then replays the sequence a chosen number of times or until it is stopped, it can even be paused.
- Hertzsprung-Russell diagram from Gaia DR3** | [↗](#)
- Queried and processed a million stars from ESA Gaia DR3 using ADQL and Python, extracting stellar temperature and luminosity data to generate a Hertzsprung-Russell diagram.
- ASCII Art Generator** | [↗](#)
- Cross-platform Rust desktop and browser image-to-text editor with luminance-based conversion, colour controls, dithering and background processing.
 - Supports .txt files and, 24-bit ANSI-colour when you want to keep the image's colour.
- Minesweeper AI Solver** | [↗](#)
- Constraint-based solver that derives safe and mine cells, enumerates valid board states, estimates mine probabilities, auto-solves deterministic moves and recognises unavoidable 50/50 guesses.
- Hypixel Skyblock Museum Tooltip Mod** | [↗](#)
- First time making a mod for Minecraft and especially for a large server like Hypixel, it is a simple mod, when you hover over an in-game item, it adds a line showing whether that item type has already been donated to your Museum.
- Terraria mod— Daybreak class extension** | [↗](#)
- First time making a mod for Terraria and inspired after the ceasing of development for the calamity mod. Added a ten-weapon progression of thrown, Daybreak-style spears. Some spears can lodge in an enemy, deal fixed damage over time, and detonate when it expires. Later weapons add themed bursts, debuffs, and secondary projectiles.

Work Experience

- The Deli Downstairs** London, UK
- Kitchen Porter — Part-time* Jul 2026 – Present
- Supported kitchen operations during high-volume service periods, maintaining clean, organised and hygienic preparation and equipment areas.
- National Student Space Conference (NSSC)** London, UK
- Volunteer* Feb 2026
- Volunteered for the conference hosted at King's College London, representing my space society for the weekend of events, and helping coordinating events and talks for speakers presenting about their expertise.
- MedMart Borehamwood** London, UK
- Pharmacist Assistant — Part-time* Jul 2023 – Sept 2024
- Sorted and handled medication for elderly patients while coordinating with colleagues to complete sensitive, time-critical tasks accurately.
- LRC - The Compton School** London, UK
- Librarian - Voluntary* Oct 2020 – June 2023
- Assisted with library organisation and rentals, student enquiries and maintaining an orderly learning environment.

Technologies

Skills: Python, Rust, C#, JS, HTML/CSS, SQL, Proficient in Linux systems, LaTeX, Pandas, API, Git, Astroquery, WebAssembly, Jupyter, Blender

Laboratory: Oscilloscope, Function Generator, Digital Multimeter